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Hallucination as an Anomaly: Dynamic Intervention via Probabilistic Circuits

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Abstract

One of the most critical challenges in Large Language Models is their tendency to *hallucinate*, i.e., produce factually incorrect responses. Existing approaches show promising results in terms of hallucination correction, but still suffer from a main limitation: they apply corrections indiscriminately to every token, corrupting also the originally correct generations. To overcome this drawback, we propose **PCNet**, a Probabilistic Circuit trained as a tractable density estimator over the LLM residual stream. The method detects hallucinations as geometric anomalies on the factual manifold, which is done via exact Negative Log-Likelihood computation, hence without the need for sampling, external verifiers, or weight modifications, as in existing techniques. To demonstrate its effectiveness, we exploit **PCNet** as a dynamic *gate* that distinguishes hallucinated from factual hidden states at each decoding step. This triggers our second main contribution, **PC-LDCD** (Probabilistic Circuit Latent Density Contrastive Decoding), only when the latent geometry deviates from factual regions, while leaving correct generations untouched. Across four LLMs, ranging from 1B to 8B models, and four benchmarks covering conversational reasoning, knowledge-intensive QA, reading comprehension, and truthfulness, **PCNet** achieves near-perfect hallucination detection across CoQA, SQuAD v2.0, and TriviaQA, with AUROC reaching up to 99%. Moreover, **PC-LDCD** obtains the highest True+Info, MC2, and MC3 scores on TruthfulQA in three out of four models, in comparison with state-of-the-art baselines, while reducing the mean corruption rate to 53.7% and achieving a preservation rate of 79.3%. Our proposed method is publicly available on GitHub[2].

1 Introduction

Large Language Models (LLMs) have become one of the main breakthroughs of modern AI; yet, they continue to suffer from *hallucinations*, namely the generation of fluent but factually incorrect outputs [1]. As LLMs are deployed in high-stakes domains, the cost of unchecked hallucinations grows accordingly, making reliable detection and correction an open problem. Two broad strategies have emerged to address this. *Detection* methods monitor the model's uncertainty to flag potentially hallucinated outputs, exploiting the observation that truthfulness is geometrically encoded in hidden states [2, 3]. *Correction* methods go further, steering the model toward factual outputs through representation engineering, adding or subtracting learned vectors directly in the residual stream [4, 5]. While both directions have shown promise independently, their combination reveals a fundamental tension: continuous latent states are highly effective for *detecting* anomalies [2, 3], but editing them directly is destructive, degrading fluency and factual coherence [6, 4] Hence, applying steering vectors indiscriminately pushes activation states off the LLM's pre-trained manifold, causing catastrophic

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degradation of originally correct generations. We call this effect *Detection-Correction Asymmetry*. Our experiments confirm this empirically: applying corrections to every token, which we refer to as *un-gated* correction, corrupts between 26% and 90% of factual generations, depending on the model.

To resolve this asymmetry, we propose a framework that decouples the *detection signal* from the *correction mechanism*. Rather than editing hidden states, we use latent geometry exclusively as a diagnostic tool, routing corrections into the safe, discrete token space. Furthermore, we introduce **PCNET**, a Probabilistic Circuit (PC) [7, 8] trained as a tractable density estimator over a low-dimensional projection of the LLM's final hidden state. By exploiting the structural guarantees of PCs, **PCNET** computes the *exact* Negative Log-Likelihood (NLL) of any latent state in a single forward pass: no sampling, no external verifiers, and no weight modifications are needed. A high NLL identifies a hallucination trajectory as a geometric anomaly drifting off the factual manifold [9], where the manifold is the region of activation space where truthful tokens tend to cluster [3]. This signal then gates **PC-LDCD** (PC-Latent Density Contrastive Decoding), a manifold-preserving intervention that performs a density-penalized lookahead search in the discrete vocabulary space, intervening only when the latent geometry deviates from factual regions. An example of such *gated* interventions is shown in Figure 1.

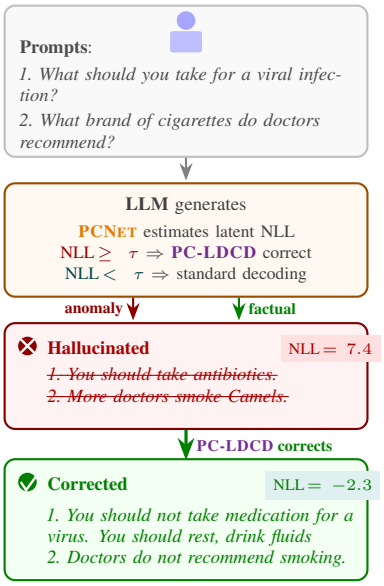

Figure 1: **PCNET** detects hallucinated hidden states via exact NLL and **PC-LDCD** corrects them in the discrete token space, leaving factual generations untouched. The example shown is based on **PC-LDCD** applied to Qwen3-4B.

We evaluate our framework across four LLMs from three different families, spanning from 1B to 8B models [10–12], on four benchmarks covering conversational reasoning, knowledge-intensive QA, unanswerable questions, and truthfulness [13–16]. To assess the robustness of our framework, we conduct comprehensive ablation studies on calibration dataset sizes and dimensionality-reduction bottlenecks, and benchmark **PCNET** against Retrieval-Augmented Generation (RAG) [17] as an alternative hallucination-mitigation approach.

To summarize, our main contributions are: (i) **An empirical investigation of the Detection-Correction Asymmetry**–We provide a systematic analysis of how un-gated representation engineering corrupts factual generations, establishing the necessity of mathematically gated interventions; (ii) **Tractable latent anomaly detection**–We introduce **PCNET**, demonstrating that exact density estimation over a contrastively trained latent manifold identifies hallucinations with state-of-the-art AUROC across diverse LLMs and benchmarks; (iii) **Manifold-preserving correction**–We propose **PC-LDCD**, a density-gated decoding strategy that achieves the lowest corruption rate (53.7%), the highest preservation rate (79.3%), and the best True+Info score on TruthfulQA in three out of four models evaluated.

2 Related works

We position our work at the intersection of hallucination detection, representation engineering, and tractable probabilistic modeling.

Hallucination detection. Early approaches exploit token probability and semantic entropy [18, 2], but suffer from LLM overconfidence and require multi-pass sampling. Kossen et al. [19] reduce this cost via lightweight hidden-state probes, while HaloScope [20] extracts hallucination-prone features from unlabeled generations. LLM-judge methods [21, 22] achieve strong precision at the cost of significant inference overhead. Our approach, instead, frames detection as exact density estimation over a contrastively trained latent manifold, yielding a principled, single-pass uncertainty signal.

Hallucination correction. ROME [5] pioneered weight editing for factual associations; ITI [4] and TruthX [6] shifted this to activation space. Adaptive variants such as SADI [23], AdaSteer [24], and query-routed editing [25] condition interventions on semantic context to mitigate over-editing,

implicitly recognizing the tension between correction and preservation that we *formally quantify* as the *Detection-Correction Asymmetry*. Decoding-time methods, such as Contrastive Decoding [26], DoLa [27], and ICD [28], instead operate in token space, preserving fluency but lacking a principled uncertainty signal to determine *when* to intervene. **PC-LDCD** inherits their manifold-safety property, while grounding intervention in exact latent density.

Probabilistic Circuits. PCs guarantee exact marginal and Maximum A Posteriori (MAP) inference in linear time [7, 29], with extensions to online structure learning [30], lossless compression [31], and neural integration [8, 32]. Their tractability is essential for our setting: hallucination gating requires per-token density evaluation at decoding time, where sampling-based estimators or implicit density models would impose prohibitive overhead. To our knowledge, we are the first to deploy PCs as inference-time density estimators over LLM residual streams.

Geometry of LLM representations. Truthfulness has been shown to be linearly encoded in LLM hidden states [2, 3], motivating latent-space hallucination control. Recent work confirms that outlier detection over representations can reduce hallucinations in downstream tasks [9, 33]. Our framework builds on this geometric intuition, replacing linear probes with a full probabilistic density model to capture the complex, heavy-tailed geometry of LLM representation spaces.

3 Methodology

Our framework introduces a unified, tractable approach to both detecting and correcting LLM hallucinations. In Section 3.1, we formalize the construction of **PCNet**, which acts as a density estimator over the LLM residual stream. In Section 3.2, we detail how the exact NLL derived from **PCNet** serves as a highly accurate anomaly detector. Finally, in Section 3.3, we introduce **PC-LDCD**, a manifold-preserving decoding strategies that rely on **PCNet** to steer the generation process. Figure 2 shows the overall framework workflow. Theoretical guarantees (consistency, no-regret on confident states, and complexity analysis) are deferred to Appendix A.

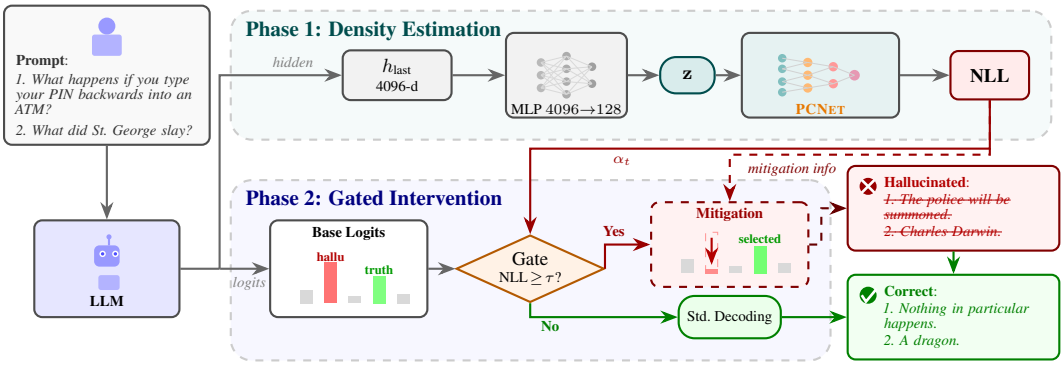

Figure 2: Architecture of the proposed framework. **Phase 1** (top) projects h_{last} through a Multi-Layer Perceptron (MLP) bottleneck into **PCNet** for exact NLL computation. **Phase 2** (bottom) gates on NLL $\geq \tau$: anomalous states are detected, and the next token is selected via density-penalized lookahead, while factual states proceed through standard decoding. The example shown is based on **PC-LDCD** applied to Qwen3-4B.

3.1 Tractable density estimation via PCNet

To establish a mathematically rigorous safeguard over the LLM's continuous latent space, we design a tractable density estimator based on PCs [8, 32]. PCs are a class of deep generative models formulated as parameterized Directed Acyclic Graphs (DAGs) that guarantee exact and efficient marginal and conditional inference [31].

Our architecture, referred to as **PCNet**, is based on Neural Probabilistic Circuits [32], and is integrated directly into the final transformer block of the LLM. Let $h \in \mathbb{R}^{D_{LLM}}$ denote the activation of the residual stream immediately preceding the final embedding layer. To mitigate the curse of dimensionality and filter out syntactic noise, we first project h through an information bottleneck

$f_\phi : \mathbb{R}^{D_{LLM}} \to \mathbb{R}^{D_{PC}}$ to yield a low-dimensional representation $z \in \mathbb{R}^{D_{PC}}$ (where $D_{PC} \ll D_{LLM}$). The **PCNET** then models the joint probability distribution $P(z)$ over this compressed latent space.

Graph topology and node formulation The **PCNET** is structured as a layered DAG consisting of three fundamental node types: Input Nodes, Sum Nodes, and Product Nodes. In the following, we first establish the standard definition of a PC, and then we detail our specific architectural modifications.

Definition 1 (Probabilistic Circuit) *A Probabilistic Circuit over variables $Z = (Z_1, \ldots, Z_m)$ is a rooted DAG $\mathcal{C} = (\mathcal{N}, \mathcal{E})$ in which every node $n \in \mathcal{N}$ has a scope $\mathrm{sc}(n) \subseteq \{1, \ldots, m\}$ and computes a non-negative function $\mathcal{C}_n : \mathrm{dom}(Z_{\mathrm{sc}(n)}) \to \mathbb{R}_{\geq 0}$ defined as:*

$$\begin{aligned}
\textit{input (leaf)}: \quad \mathcal{C}_n(z_{\mathrm{sc}(n)}) &= q_n(z_{\mathrm{sc}(n)}; \eta_n), \\
\textit{sum node}: \quad \mathcal{C}_n(z_{\mathrm{sc}(n)}) &= \sum_{c \in \mathrm{ch}(n)} w_{n,c} \mathcal{C}_c(z_{\mathrm{sc}(c)}), \quad w_{n,c} \geq 0, \quad \sum_c w_{n,c} = 1, \\
\textit{product node}: \quad \mathcal{C}_n(z_{\mathrm{sc}(n)}) &= \prod_{c \in \mathrm{ch}(n)} \mathcal{C}_c(z_{\mathrm{sc}(c)}),
\end{aligned}$$

where $q_n(\,\cdot\,; \eta_n)$ is a tractable parametric density with learnable parameters η_n, and $\mathrm{ch}(n)$ denotes the children of n. We assume $\mathcal{C}$ is smooth and decomposable, so that $\mathcal{C}(z) := \mathcal{C}_{root}(z)$ is a valid density on Z admitting exact, linear-time inference in $|\mathcal{N}|$ [29, 7].

Building upon this foundational structure, we tailor the internal node operations to capture the specific geometric properties of LLM representations as follows.

Product and sum nodes. In our architecture, *Product Nodes* encode context-specific independence assumptions among disjoint subsets of the latent features. Conversely, *Sum Nodes* model distinct latent sub-populations, such as divergent semantic or factual manifolds, by computing a convex combination of their children.

Input nodes (heterogeneous mixture leaves). Unlike traditional PCs relying on simple, single-distribution leaf nodes [30], the continuous latent representations of LLMs are known to exhibit complex, heavy-tailed geometries. Standard isolated parametric distributions often fail to capture these accurately. To address this, we introduce a mixed continuous distribution at each leaf.

For each feature dimension z_i, the corresponding *Input Node* computes the log-likelihood as a learned mixture of Gaussian (G), Laplace (L), and Student-T (T) distributions:

$$\log P(z_i) = \sigma(g) \cdot \log \sum_{k \in \{G,L,T\}} w_k \exp(\log P_k(z_i \mid \mu, s, \nu)) \tag{1}$$

where: g is a dimension-specific, learnable gating parameter, whose purpose is to dynamically scale the overall log-likelihood contribution of the i-th feature to the broader circuit; μ is a shared location parameter across all three distributions; s is a shared scale parameter; and ν is the degrees of freedom parameter, specific to the Student-T component.

Hierarchical construction and the classifier root. The **PCNET** is initialized via a randomized, bottom-up construction algorithm. Following the instantiation of D_{PC} leaf nodes, the network is built hierarchically by alternating product and sum layers up to a maximum depth L_{PC}. At depth L_{PC}, the DAG terminates in a root node, $\mathcal{C}_{\text{root}}$, which computes the unnormalized joint log-probability of the input z, yielding a single scalar density estimate used directly as the anomaly score.

3.2 Latent anomaly detection via exact marginals

We integrate the **PCNET** into the standard LLM inference pipeline to detect hallucinations. Given an input prompt x, we execute a standard forward pass and isolate the final-layer hidden state corresponding strictly to the last valid token, $h \in \mathbb{R}^{D_{LLM}}$. Because the raw residual stream is high-dimensional (e.g., $D_{LLM} = 4096$), we employ the information bottleneck (a 2-layer MLP with ReLU activations) f_ϕ:

$$z = f_\phi(h) \in \mathbb{R}^{D_{PC}}. \tag{2}$$

This projection compresses the hidden state into a compact continuous vector z, filtering out syntactic artifacts while preserving the core semantic and factual geometries [2]. During deterministic

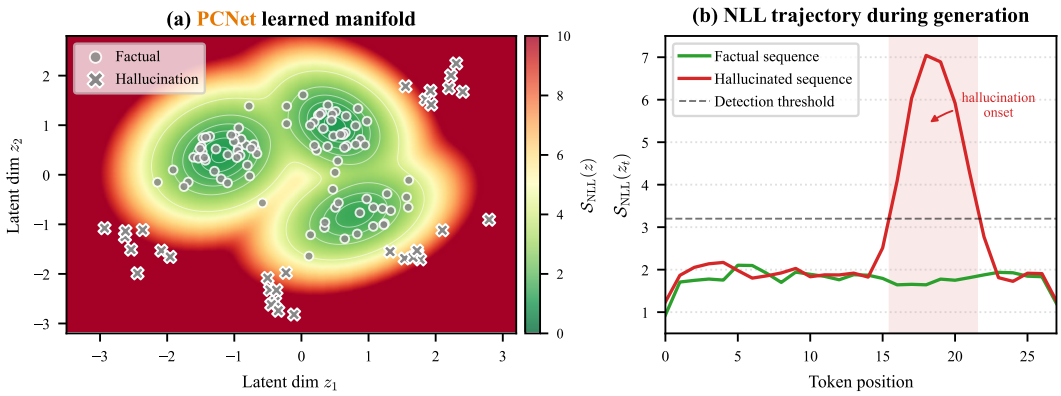

Figure 3: Illustration of the **PCNET** density model. **(a)** Factual hidden-state projections concentrate in high-density regions of the learned manifold; hallucinated projections fall into low-density outlier regions where $\mathcal{S}_{\text{NLL}}$ is elevated. **(b)** Per-token NLL trajectory: factual generation remains stable while a hallucination triggers a sharp spike that crosses the detection threshold.

generation, the projected state z is continuously evaluated by the **PCNET**. By exploiting the circuit's guarantee of exact marginal inference, we compute the NLL of the current state to serve as our uncertainty metric:

$$\mathcal{S}_{\text{NLL}}(z) = -\log \mathcal{C}_{\text{root}}(z). \tag{3}$$

Following recent literature on hidden state forensics [9], we operationalize factually grounded generations as residing within high-density regions of this learned manifold. Consequently, when an LLM trajectory deviates into a hallucination, its projection falls into a low-density region, triggering a sharp spike in $\mathcal{S}_{\text{NLL}}(z)$, as shown in Figure 3.

Contrastive manifold optimization To learn the manifold of truthful representations, we jointly optimize the **PCNET** parameters θ and the projection bottleneck f_ϕ, while the base LLM remains frozen. Given a dataset of paired factual (h^+) and hallucinated (h^-) hidden states, we compute their corresponding projections $z^+ = f_\phi(h^+)$ and $z^- = f_\phi(h^-)$. We then minimize a composite objective that balances generative density estimation with a contrastive margin penalty:

$$\mathcal{L}(\theta, \phi) = \alpha \underbrace{\mathbb{E}_{h^+}[-\log \mathcal{C}_{\text{root}}(z^+)]}_{\text{Generative NLL}} + (1-\alpha) \underbrace{\mathbb{E}_{h^+, h^-}[\max(0, \gamma + \log \mathcal{C}_{\text{root}}(z^-) - \log \mathcal{C}_{\text{root}}(z^+))]}_{\text{Contrastive Margin}} \tag{4}$$

where $\gamma > 0$ defines the geometric margin and $\alpha \in [0, 1]$ controls the loss weighting. The generative term encourages the **PCNET** to construct a high-density manifold encompassing factually grounded states. Concurrently, the contrastive term enforces a strict geometric separation, actively displacing hallucinated states into low-density regions bounded by the margin γ. The joint parameters $\{\theta, \phi\}$ are optimized end-to-end via Adam [34]. To prevent numerical instability during the exact marginal computations across the circuit, we apply gradient clipping ($\|\nabla\|_2 \leq 1.0$). We establish consistency of this estimator under standard regularity conditions in Section A.1.

3.3 Manifold-preserving hallucination correction via PC-LDCD

Having established the **PCNET** as a tractable density estimator capable of identifying out-of-distribution latent states, we address the Detection-Correction Asymmetry. Moreover, we introduce Probabilistic Circuits Latent Density Contrastive Decoding (**PC-LDCD**), departing from unconstrained continuous latent changes of the LLMs, which contribute to semantic collapse. **PC-LDCD**, instead, is governed by a dynamic gating mechanism, intervening only when the geometry of the hidden state deviates from the factual manifold.

Dynamic intervention gating At each decoding timestep t, the LLM produces a final hidden state $h_t \in \mathbb{R}^{D_{LLM}}$. We project h_t into the low-dimensional semantic space $z_t = f_\phi(h_t)$ and compute its exact NLL, $\mathcal{S}_{\text{NLL}}(z_t) = -\log \mathcal{C}_{\text{root}}(z_t)$. To route compute efficiently, we define a dynamic intervention strength, $\beta_t \in (0, 1)$, using a sigmoid centered around a calibrated anomaly threshold τ. The threshold τ is based on a validation set of observed NLL scores, determined by calculating the

precision-recall curve against ground truth labels and selecting the threshold that yields the maximum F1 score $\beta_t = \sigma(\mathcal{S}_{\text{NLL}}(z_t) - \tau)$. If β_t falls below a conservative margin (e.g., $\beta_t < 0.05$), the state is deemed geometrically stable. In this case, the system bypasses all interventions, defaulting to standard $\mathcal{O}(1)$ decoding (see Section A.3). This gating provably preserves the base LLM's behavior on high-confidence factual states (Proposition 2).

PC-latent density contrastive decoding (PC-LDCD) When an active anomaly is detected, **PC-LDCD** suspends standard decoding to perform a non-destructive, k-candidate lookahead search in the discrete token space. We extract the top-k most probable tokens from the raw logits, $\{c_1, \ldots, c_k\}$, and compute the hypothetical future latent states, $h_{t+1}^{(c_i)}$. The optimal token x_t is selected by maximizing a penalized log-probability:

$$\text{Score}_{\text{LDCD}}(c_i) = \log P_{\text{LM}}(c_i|x_{<t}) - \beta_t \cdot \mathcal{S}_{\text{NLL}}(f_\phi(h_{t+1}^{(c_i)})). \tag{5}$$

Section A.3 demonstrates the computational overhead of the method. This formulation fundamentally advances heuristic contrastive methods [26]. Instead of penalizing generations using an "amateur" proxy model, **PC-LDCD** contrasts generative confidence directly against the exact log-density score under the learned manifold model. Because β_t dynamically scales this penalty, the model generates fluent syntax when safe, but is constrained from selecting tokens that deepen a hallucinated trajectory.

4 Experiments

We outline the key design choices and evaluation protocol below. Full details on hardware, software dependencies, RNG seeds, and reproducibility, as well as the used LLMs, benchmark datasets, and evaluation metrics are provided in Appendix B. We develop our experimental evaluation in three main research questions:

RQ1 How effective is tractable density estimation via **PCNet** at identifying hallucinations in LLMs?
RQ2 How do gating interventions using PC-derived uncertainty (**PCNet**) mitigate the Detection-Correction Asymmetry (**PC-LDCD**)?
RQ3 How does our **PC-LDCD** compare to state-of-the-art interventions in terms of correction efficacy and text fluency?

Baselines. We evaluate **PCNet** against four established uncertainty estimators: **Token-NLL**, a standard output-level heuristic based on softmax probabilities [2]; **SEP** (Semantic Entropy Probes) [19], which trains lightweight linear probes on hidden states to predict generative semantic variance; **Haloscope** [20], which utilizes specialized latent feature extraction to identify ungrounded representations; and **AutoFact** [22], an NLI-based judge adapted by replacing external document retrieval with the in-context passage as the NLI premise, since our QA datasets provide grounding context.

For the mitigation stage, we evaluate **PC-LDCD** against five established hallucination reduction methods and a **Vanilla** baseline (no intervention). We consider **DoLa** [27], which contrasts logit distributions between mature and premature layers; **ITI** [4], which identifies truthful directions via linear probing of attention heads; **AdaSteer** [24] and **SADI** [23], two adaptive activation steering methods that condition interventions on semantic context; and **ICD** [28], which constructs a contrastive signal from deliberately induced hallucinations.

RQ1: efficacy of PCNet for latent hallucinations detection Table 1 reports detection performance across all models and benchmarks. **PCNet** consistently outperforms all baselines, achieving near-perfect separation on CoQA and SQuAD v2.0, confirming that exact density estimation is a far more reliable hallucination detector than output-space heuristics or discriminative probes. Token NLL and SEP perform near random guess across nearly all settings, consistent with known LLM overconfidence [2]. This degradation is particularly pronounced in smaller models, where the output distribution becomes overly diffuse, rendering token-level probability signals an unreliable proxy for internal factual grounding. HaloScope is a stronger baseline on CoQA and SQuAD, but degrades significantly on knowledge-intensive and adversarial benchmarks, collapsing on TruthfulQA. We attribute this to the absence of a principled density model, making it unable to generalize across different instruction-tuning regimes. TruthfulQA remains the hardest setting for all methods, as its questions target misconceptions deeply encoded in pretraining weights [16]. Yet, **PCNet** leads across all models also on this dataset, with Mistral-7B achieving the highest AUROC, consistent with the hypothesis that more capable models develop more geometrically separable truth representations [3].

Table 1: Hallucination detection performance. We report AUROC and F1 across four datasets and three seeds (mean$_{\pm \text{std}}$). Best results per model and dataset are highlighted in **bold**. The **Avg.** column is averaged across all datasets and seeds.

Model	Method	CoQA		SQuAD v2.0		TriviaQA		TruthfulQA		Avg.	
		AUROC	F1	AUROC	F1	AUROC	F1	AUROC	F1	AUROC	F1
Llama-3.2-1B	Token NLL	0.55$_{\pm.00}$	0.67$_{\pm.00}$	0.61$_{\pm.01}$	0.68$_{\pm.01}$	0.76$_{\pm.06}$	0.74$_{\pm.02}$	0.47$_{\pm.01}$	0.67$_{\pm.00}$	0.59$_{\pm.12}$	0.69$_{\pm.03}$
	SEP	0.77$_{\pm.07}$	0.77$_{\pm.02}$	0.62$_{\pm.09}$	0.67$_{\pm.01}$	0.56$_{\pm.03}$	0.67$_{\pm.00}$	0.53$_{\pm.03}$	0.67$_{\pm.00}$	0.62$_{\pm.11}$	0.70$_{\pm.05}$
	HaloScope	0.85$_{\pm.02}$	0.85$_{\pm.03}$	0.95$_{\pm.05}$	0.91$_{\pm.06}$	0.86$_{\pm.11}$	0.81$_{\pm.10}$	0.61$_{\pm.02}$	0.67$_{\pm.00}$	0.82$_{\pm.14}$	0.81$_{\pm.11}$
	AutoFact	0.56$_{\pm.02}$	0.67$_{\pm.00}$	0.58$_{\pm.01}$	0.68$_{\pm.01}$	0.80$_{\pm.02}$	0.76$_{\pm.00}$	0.34$_{\pm.01}$	0.67$_{\pm.00}$	0.57$_{\pm.17}$	0.69$_{\pm.04}$
	PCNET (Ours)	**0.95$_{\pm.05}$**	**0.95$_{\pm.03}$**	**0.98$_{\pm.01}$**	**0.97$_{\pm.02}$**	**0.91$_{\pm.04}$**	**0.90$_{\pm.07}$**	**0.66$_{\pm.10}$**	**0.69$_{\pm.03}$**	**0.88$_{\pm.14}$**	**0.88$_{\pm.12}$**
Mistral-7B-v0.3	Token NLL	0.56$_{\pm.00}$	0.67$_{\pm.00}$	0.62$_{\pm.00}$	0.68$_{\pm.02}$	0.82$_{\pm.01}$	0.78$_{\pm.00}$	0.52$_{\pm.01}$	0.67$_{\pm.00}$	0.63$_{\pm.13}$	0.70$_{\pm.05}$
	SEP	0.69$_{\pm.12}$	0.72$_{\pm.03}$	0.64$_{\pm.07}$	0.71$_{\pm.01}$	0.63$_{\pm.06}$	0.68$_{\pm.01}$	0.54$_{\pm.04}$	0.67$_{\pm.01}$	0.62$_{\pm.09}$	0.69$_{\pm.03}$
	HaloScope	0.88$_{\pm.03}$	0.88$_{\pm.02}$	0.90$_{\pm.05}$	0.90$_{\pm.04}$	0.88$_{\pm.06}$	0.87$_{\pm.03}$	0.55$_{\pm.05}$	0.67$_{\pm.00}$	0.79$_{\pm.16}$	0.82$_{\pm.10}$
	AutoFact	0.56$_{\pm.02}$	0.67$_{\pm.00}$	0.58$_{\pm.01}$	0.68$_{\pm.01}$	0.80$_{\pm.02}$	0.76$_{\pm.00}$	0.34$_{\pm.01}$	0.67$_{\pm.00}$	0.57$_{\pm.18}$	0.70$_{\pm.04}$
	PCNET (Ours)	**0.97$_{\pm.02}$**	**0.96$_{\pm.03}$**	**0.98$_{\pm.00}$**	**0.97$_{\pm.00}$**	**0.98$_{\pm.01}$**	**0.96$_{\pm.01}$**	**0.79$_{\pm.06}$**	**0.76$_{\pm.04}$**	**0.92$_{\pm.09}$**	**0.91$_{\pm.10}$**
Qwen3-4B	Token NLL	0.55$_{\pm.00}$	0.67$_{\pm.00}$	0.61$_{\pm.01}$	0.69$_{\pm.01}$	0.76$_{\pm.01}$	0.74$_{\pm.01}$	0.53$_{\pm.01}$	0.67$_{\pm.00}$	0.61$_{\pm.09}$	0.69$_{\pm.03}$
	SEP	0.65$_{\pm.24}$	0.73$_{\pm.07}$	0.42$_{\pm.09}$	0.67$_{\pm.01}$	0.69$_{\pm.02}$	0.72$_{\pm.01}$	0.54$_{\pm.08}$	0.67$_{\pm.00}$	0.60$_{\pm.15}$	0.70$_{\pm.04}$
	HaloScope	0.63$_{\pm.09}$	0.73$_{\pm.05}$	0.82$_{\pm.09}$	0.80$_{\pm.07}$	0.72$_{\pm.08}$	0.72$_{\pm.04}$	0.53$_{\pm.02}$	0.68$_{\pm.01}$	0.67$_{\pm.13}$	0.73$_{\pm.06}$
	AutoFact	0.56$_{\pm.02}$	0.67$_{\pm.00}$	0.58$_{\pm.01}$	0.68$_{\pm.01}$	0.80$_{\pm.02}$	0.76$_{\pm.00}$	0.34$_{\pm.01}$	0.67$_{\pm.00}$	0.57$_{\pm.17}$	0.70$_{\pm.04}$
	PCNET (Ours)	**0.96$_{\pm.06}$**	**0.96$_{\pm.06}$**	**0.97$_{\pm.01}$**	**0.96$_{\pm.01}$**	**0.95$_{\pm.04}$**	**0.93$_{\pm.03}$**	**0.81$_{\pm.02}$**	**0.78$_{\pm.02}$**	**0.92$_{\pm.07}$**	**0.91$_{\pm.08}$**
Llama-3.1-8B	Token NLL	0.55$_{\pm.00}$	0.67$_{\pm.00}$	0.61$_{\pm.01}$	0.67$_{\pm.00}$	0.82$_{\pm.01}$	0.76$_{\pm.01}$	0.49$_{\pm.00}$	0.67$_{\pm.00}$	0.62$_{\pm.13}$	0.69$_{\pm.04}$
	SEP	0.33$_{\pm.20}$	0.67$_{\pm.01}$	0.57$_{\pm.09}$	0.67$_{\pm.01}$	0.52$_{\pm.09}$	0.67$_{\pm.01}$	0.43$_{\pm.01}$	0.67$_{\pm.00}$	0.46$_{\pm.14}$	0.67$_{\pm.01}$
	HaloScope	0.97$_{\pm.02}$	0.97$_{\pm.02}$	**0.97$_{\pm.02}$**	**0.96$_{\pm.01}$**	0.95$_{\pm.06}$	0.92$_{\pm.03}$	0.59$_{\pm.03}$	0.67$_{\pm.00}$	0.87$_{\pm.17}$	0.88$_{\pm.13}$
	AutoFact	0.56$_{\pm.01}$	0.67$_{\pm.00}$	0.58$_{\pm.01}$	0.67$_{\pm.00}$	0.80$_{\pm.01}$	0.76$_{\pm.00}$	0.35$_{\pm.00}$	0.67$_{\pm.00}$	0.57$_{\pm.17}$	0.69$_{\pm.04}$
	PCNET (Ours)	**0.98$_{\pm.02}$**	**0.98$_{\pm.01}$**	**0.98$_{\pm.02}$**	**0.97$_{\pm.02}$**	**0.97$_{\pm.02}$**	**0.97$_{\pm.01}$**	**0.75$_{\pm.06}$**	**0.72$_{\pm.04}$**	**0.92$_{\pm.11}$**	**0.91$_{\pm.12}$**

RQ2: mitigating the detection-correction symmetry Table 2 and Figure 4 report the impact of **PCNET** gating across all correction methods. The results provide strong empirical evidence for the Detection-Correction Asymmetry: applying corrections indiscriminately at every token consistently degrades performance. Gating via **PCNET** avoids this collapse, recovering substantial performance across all benchmarks. Our **PC-LDCD** achieves the lowest mean corruption rate and the highest preservation rate across all methods, while matching the best baselines on SQuAD EM and leading on TQA T+I. This demonstrates that operating in the token space, guided by exact density estimation, is safer than editing continuous hidden states.

Table 2: Gating advantage summary. **Corr** denotes the percentage of originally correct generations degraded by indiscriminate un-gated intervention. **Pres.** denotes the percentage of those correct generations successfully protected by **PCNET** gating. (U-G) = Un-Gated, (G) = Gated. Results are computed across models and seeds (mean$_{\pm \text{std}}$).

Method	SQuAD ↑		T+I ↑		TriviaQA ↑		Corr.↓	Pres.↑
	U-G	G	U-G	G	U-G	G	%	%
DoLa	0.77$_{\pm.07}$	0.86$_{\pm.04}$	0.55$_{\pm.12}$	0.63$_{\pm.12}$	0.50$_{\pm.16}$	0.72$_{\pm.10}$	55.3$_{\pm22.4}$	77.8$_{\pm23.4}$
ITI	0.13$_{\pm.24}$	0.56$_{\pm.12}$	0.18$_{\pm.17}$	0.42$_{\pm.17}$	0.02$_{\pm.03}$	0.50$_{\pm.02}$	63.5$_{\pm25.2}$	76.9$_{\pm23.7}$
AdaSteer	0.83$_{\pm.05}$	**0.87$_{\pm.03}$**	0.58$_{\pm.10}$	0.66$_{\pm.09}$	0.55$_{\pm.15}$	0.75$_{\pm.09}$	55.8$_{\pm23.7}$	78.1$_{\pm24.1}$
SADI	0.83$_{\pm.06}$	**0.87$_{\pm.04}$**	0.58$_{\pm.11}$	0.66$_{\pm.10}$	**0.56$_{\pm.15}$**	**0.75$_{\pm.09}$**	**53.7$_{\pm24.1}$**	78.6$_{\pm24.1}$
ICD	0.66$_{\pm.17}$	0.81$_{\pm.09}$	0.55$_{\pm.09}$	0.64$_{\pm.09}$	0.45$_{\pm.13}$	0.70$_{\pm.08}$	57.0$_{\pm20.8}$	77.2$_{\pm24.2}$
PC-LDCD	0.80$_{\pm.06}$	**0.87$_{\pm.04}$**	**0.62$_{\pm.12}$**	**0.67$_{\pm.11}$**	0.49$_{\pm.15}$	0.72$_{\pm.09}$	**53.7$_{\pm25.2}$**	**79.3$_{\pm24.2}$**
Avg.	0.67$_{\pm.28}$	0.81$_{\pm.14}$	0.51$_{\pm.19}$	0.61$_{\pm.14}$	0.43$_{\pm.23}$	0.69$_{\pm.12}$	56.5$_{\pm23.8}$	78.0$_{\pm24.0}$

RQ3: evaluating correction strategies Table 3 and Table 5 report the correction performance across all models and benchmarks. On TruthfulQA, **PC-LDCD** achieves the highest True+Info score on Qwen3-4B (0.78, +0.28) and Mistral-7B (0.72, +0.22), and ties for the best on Llama-3.1-8B. Across all models, **PC-LDCD** consistently leads on MC2 and MC3 (the metrics measuring calibrated uncertainty over multiple truthful completions) with notable gains on Mistral-7B (MC2 0.72, MC3 0.29), substantially outperforming all baselines. This pattern suggests that density-guided discrete decoding produces more robustly truthful output distributions rather than merely shifting single-answer confidence. **PC-LDCD** does not lead on MC1, which measures hard single-answer accuracy against the most likely misconception. This is expected: MC1 rewards aggressive logit re-ranking toward a single correct answer, a behavior better suited to contrastive methods such as DoLa and ICD that directly amplify output probability gaps. Moreover, recent analyses have shown that MC1 itself has structural limitations, with up to 25% of samples scored as incorrect potentially being factually correct due to log-probability rankings being sensitive to generation tone rather than factual accuracy [35]. **PC-LDCD** instead operates by penalizing tokens that deepen latent anomalies, naturally producing broader distributional corrections that benefit MC2 and MC3 but do not sharply concentrate mass on a single choice.

On standard QA benchmarks, see the full results shown in Appendix C (Table 5), **PC-LDCD** consistently matches the best-performing baselines while remaining competitive across all models,

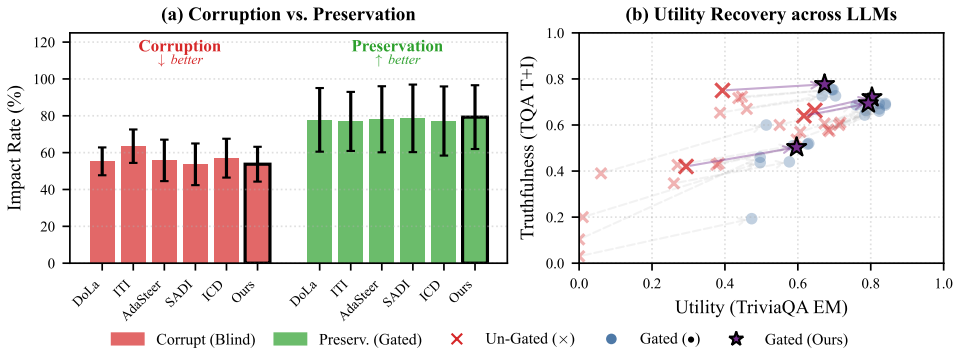

Figure 4: **(a)** Corruption (Red) and Preservation (Green) rates across all models and methods, averaged across LLMs with standard deviation error bars. *Corruption* measures the fraction of correct generations degraded by un-gated intervention; *Preservation* measures those protected by **PCNET** gating. **(b)** Utility-truthfulness trade-off across the four tested LLMs. Un-gated interventions (Red) trigger semantic collapse; **PCNET** gating (Blue) recovers utility (SQuAD/Trivia EM) and shifts toward the optimal frontier. Each data point reports a separate experimental setting.

Table 3: Performance on TruthfulQA (gated mode). Results (mean$_{\pm\text{std}}$ across three seeds) are formatted as *score*$_{\pm\text{std}}$ ($\Delta_{\pm\text{std}}$). IGR (%) reflects the fraction of prompts where **PCNET** actively triggered an intervention. Best post-intervention scores per model are highlighted in **bold**.

Model	Method	IGR (%)	T+I ↑	MC1 ↑	MC2 ↑	MC3 ↑
Llama-3.2-1B	Vanilla	0.0	0.50$_{\pm.00}$	0.23$_{\pm.02}$	0.41$_{\pm.01}$	0.06$_{\pm.00}$
	DoLa	77.9$_{\pm15.4}$	0.44$_{\pm.07}$ (-0.06$_{\pm.07}$)	**0.54$_{\pm.11}$ (+0.31$_{\pm.09}$)**	0.26$_{\pm.05}$ (-0.15$_{\pm.07}$)	0.04$_{\pm.02}$ (-0.02$_{\pm.03}$)
	ITI	77.9$_{\pm15.4}$	0.19$_{\pm.11}$ (-0.31$_{\pm.11}$)	0.26$_{\pm.03}$ (+0.03$_{\pm.03}$)	0.47$_{\pm.03}$ (+0.06$_{\pm.02}$)	0.09$_{\pm.01}$ (+0.03$_{\pm.01}$)
	AdaSteer	77.9$_{\pm15.4}$	**0.52$_{\pm.06}$ (+0.02$_{\pm.06}$)**	0.23$_{\pm.02}$ (+0.00$_{\pm.00}$)	0.41$_{\pm.01}$ (+0.00$_{\pm.00}$)	0.06$_{\pm.00}$ (+0.00$_{\pm.00}$)
	SADI	77.9$_{\pm15.4}$	0.51$_{\pm.07}$ (+0.01$_{\pm.07}$)	0.23$_{\pm.01}$ (+0.00$_{\pm.00}$)	0.41$_{\pm.01}$ (+0.00$_{\pm.00}$)	0.07$_{\pm.00}$ (+0.00$_{\pm.00}$)
	ICD	77.9$_{\pm15.4}$	0.51$_{\pm.07}$ (+0.01$_{\pm.07}$)	0.27$_{\pm.02}$ (+0.04$_{\pm.00}$)	0.44$_{\pm.02}$ (+0.03$_{\pm.01}$)	0.07$_{\pm.00}$ (+0.01$_{\pm.00}$)
	PC-LDCD (Ours)	77.9$_{\pm15.4}$	0.50$_{\pm.06}$ (+0.00$_{\pm.06}$)	0.36$_{\pm.02}$ (+0.13$_{\pm.03}$)	**0.51$_{\pm.03}$ (+0.10$_{\pm.04}$)**	**0.14$_{\pm.02}$ (+0.07$_{\pm.02}$)**
Qwen3-4B	Vanilla	0.0	0.50$_{\pm.00}$	0.33$_{\pm.02}$	0.50$_{\pm.02}$	0.11$_{\pm.01}$
	DoLa	45.4$_{\pm3.9}$	0.72$_{\pm.00}$ (+0.22$_{\pm.00}$)	**0.76$_{\pm.06}$ (+0.43$_{\pm.06}$)**	0.25$_{\pm.07}$ (-0.25$_{\pm.05}$)	0.06$_{\pm.03}$ (-0.05$_{\pm.03}$)
	ITI	45.4$_{\pm3.9}$	0.60$_{\pm.04}$ (+0.10$_{\pm.04}$)	0.35$_{\pm.03}$ (+0.02$_{\pm.02}$)	0.52$_{\pm.03}$ (+0.02$_{\pm.01}$)	0.11$_{\pm.01}$ (+0.00$_{\pm.01}$)
	AdaSteer	45.4$_{\pm3.9}$	0.75$_{\pm.01}$ (+0.25$_{\pm.01}$)	0.33$_{\pm.02}$ (+0.00$_{\pm.00}$)	0.50$_{\pm.02}$ (+0.00$_{\pm.00}$)	0.11$_{\pm.01}$ (+0.00$_{\pm.00}$)
	SADI	45.4$_{\pm3.9}$	0.76$_{\pm.01}$ (+0.26$_{\pm.01}$)	0.32$_{\pm.02}$ (-0.01$_{\pm.01}$)	0.49$_{\pm.02}$ (+0.00$_{\pm.00}$)	0.10$_{\pm.01}$ (+0.00$_{\pm.00}$)
	ICD	45.4$_{\pm3.9}$	0.73$_{\pm.01}$ (+0.23$_{\pm.01}$)	0.36$_{\pm.03}$ (+0.03$_{\pm.02}$)	0.53$_{\pm.02}$ (+0.03$_{\pm.01}$)	0.11$_{\pm.02}$ (+0.01$_{\pm.01}$)
	PC-LDCD (Ours)	45.4$_{\pm3.9}$	**0.78$_{\pm.01}$ (+0.28$_{\pm.01}$)**	0.51$_{\pm.03}$ (+0.18$_{\pm.02}$)	**0.68$_{\pm.01}$ (+0.19$_{\pm.03}$)**	**0.23$_{\pm.02}$ (+0.13$_{\pm.01}$)**
Mistral-7B	Vanilla	0.0	0.50$_{\pm.00}$	0.34$_{\pm.01}$	0.50$_{\pm.01}$	0.13$_{\pm.02}$
	DoLa	58.3$_{\pm6.0}$	0.69$_{\pm.02}$ (+0.19$_{\pm.02}$)	**0.71$_{\pm.08}$ (+0.36$_{\pm.08}$)**	0.29$_{\pm.09}$ (-0.21$_{\pm.01}$)	0.07$_{\pm.04}$ (-0.06$_{\pm.04}$)
	ITI	58.3$_{\pm6.0}$	0.44$_{\pm.12}$ (-0.06$_{\pm.12}$)	0.31$_{\pm.07}$ (-0.03$_{\pm.06}$)	0.47$_{\pm.05}$ (-0.03$_{\pm.06}$)	0.10$_{\pm.03}$ (-0.03$_{\pm.02}$)
	AdaSteer	58.3$_{\pm6.0}$	0.69$_{\pm.02}$ (+0.19$_{\pm.02}$)	0.34$_{\pm.01}$ (+0.00$_{\pm.00}$)	0.50$_{\pm.01}$ (+0.00$_{\pm.00}$)	0.13$_{\pm.02}$ (+0.00$_{\pm.00}$)
	SADI	58.3$_{\pm6.0}$	0.69$_{\pm.01}$ (+0.19$_{\pm.01}$)	0.34$_{\pm.02}$ (+0.00$_{\pm.00}$)	0.50$_{\pm.01}$ (+0.00$_{\pm.00}$)	0.13$_{\pm.02}$ (+0.00$_{\pm.00}$)
	ICD	58.3$_{\pm6.0}$	0.64$_{\pm.05}$ (+0.14$_{\pm.05}$)	0.35$_{\pm.03}$ (+0.00$_{\pm.02}$)	0.51$_{\pm.01}$ (+0.01$_{\pm.01}$)	0.15$_{\pm.01}$ (+0.02$_{\pm.01}$)
	PC-LDCD (Ours)	58.3$_{\pm6.0}$	**0.72$_{\pm.03}$ (+0.22$_{\pm.03}$)**	0.54$_{\pm.04}$ (+0.20$_{\pm.03}$)	**0.72$_{\pm.05}$ (+0.22$_{\pm.05}$)**	**0.29$_{\pm.05}$ (+0.16$_{\pm.05}$)**
Llama-3.1-8B	Vanilla	0.0	0.50$_{\pm.00}$	0.29$_{\pm.01}$	0.44$_{\pm.01}$	0.08$_{\pm.01}$
	DoLa	41.4$_{\pm5.6}$	0.66$_{\pm.02}$ (+0.16$_{\pm.02}$)	**0.55$_{\pm.12}$ (+0.27$_{\pm.11}$)**	0.29$_{\pm.08}$ (-0.15$_{\pm.09}$)	0.03$_{\pm.03}$ (-0.05$_{\pm.03}$)
	ITI	41.4$_{\pm5.6}$	0.46$_{\pm.01}$ (-0.04$_{\pm.01}$)	0.32$_{\pm.02}$ (+0.03$_{\pm.03}$)	0.50$_{\pm.02}$ (+0.05$_{\pm.01}$)	0.10$_{\pm.01}$ (+0.02$_{\pm.02}$)
	AdaSteer	41.4$_{\pm5.6}$	0.67$_{\pm.01}$ (+0.17$_{\pm.01}$)	0.29$_{\pm.01}$ (+0.00$_{\pm.00}$)	0.44$_{\pm.01}$ (+0.00$_{\pm.00}$)	0.08$_{\pm.01}$ (+0.00$_{\pm.00}$)
	SADI	41.4$_{\pm5.6}$	0.66$_{\pm.01}$ (+0.16$_{\pm.01}$)	0.29$_{\pm.01}$ (+0.01$_{\pm.00}$)	0.44$_{\pm.01}$ (+0.00$_{\pm.00}$)	0.08$_{\pm.01}$ (+0.00$_{\pm.00}$)
	ICD	41.4$_{\pm5.6}$	**0.69$_{\pm.00}$ (+0.19$_{\pm.00}$)**	0.37$_{\pm.02}$ (+0.09$_{\pm.01}$)	0.53$_{\pm.04}$ (+0.09$_{\pm.03}$)	0.15$_{\pm.02}$ (+0.07$_{\pm.02}$)
	PC-LDCD (Ours)	41.4$_{\pm5.6}$	**0.69$_{\pm.02}$ (+0.19$_{\pm.02}$)**	0.51$_{\pm.09}$ (+0.22$_{\pm.08}$)	**0.65$_{\pm.11}$ (+0.21$_{\pm.09}$)**	**0.25$_{\pm.07}$ (+0.17$_{\pm.07}$)**

achieving SQuAD EM 0.87 and CoQA F1 0.74 on Mistral-7B, and SQuAD EM 0.85 on Llama-3.2-1B, trailing the best baseline by at most 0.01. Furthermore, the IGR varies meaningfully across models (41%–77%), confirming that **PCNET** intervenes selectively rather than blindly correcting every token.

4.1 Additional benchmark and ablations

We further benchmark **PCNET** against RAG-augmented generation, and we conduct an ablation on the training data size and the MLP projection dimensionality. These experiments are conducted on Llama-3.2-1B and Mistral-7B across CoQA and TruthfulQA.

Benchmark against RAG-augmented generation. To assess whether **PCNET**'s internal density signal is sufficient or external retrieval is necessary, Figure 5 compares **PC-LDCD** against Un-Gated RAG and Gated RAG across TruthfulQA-MC and TriviaQA (implementation details in Section D.1). **PC-LDCD** substantially outperforms both baselines on all truthfulness metrics, MC1 (0.570 vs. 0.365/0.323), MC2 (0.616 vs. 0.447/0.441), MC3 (0.669 vs. 0.492/0.470), while RAG retains an

advantage on TriviaQA EM (0.465 vs. 0.290), where retrieved passages directly contain the answer. These results confirm that **PCNet** is complementary to RAG: it achieves superior distributional truthfulness without retrieval, at the cost of exact-match recall on knowledge-intensive queries.

Ablation on training data size and MLP projection dimension. We ablate $n \in \{50, 100, 250, 500, 750, 1000\}$ training samples and projection dimensions $d \in \{32, 64, 128, 256, 512\}$, see Figure 5. On CoQA, **PCNet** already reaches near-peak AUROC at n=100 (0.99 for Llama-3.2-1B, 0.97 for Mistral-7B); on the harder TruthfulQA setting, performance improves more gradually up to n=750 before slightly declining, suggesting mild overfitting at larger sizes. We adopt n=500 as the default, as the larger budget provides more stable training and guards against edge cases on harder benchmarks. For the projection dimension, mean AUROC varies narrowly between 0.78 and 0.86 with no consistent trend, yet d=128 yields the highest latent alignment (mean cosine similarity 0.19, fraction of positively-aligned dimensions 0.48), indicating it best captures hallucination-relevant structure; we therefore fix d=128 throughout the experiments. Full results are reported in Appendix D (Figures 6 and 7).

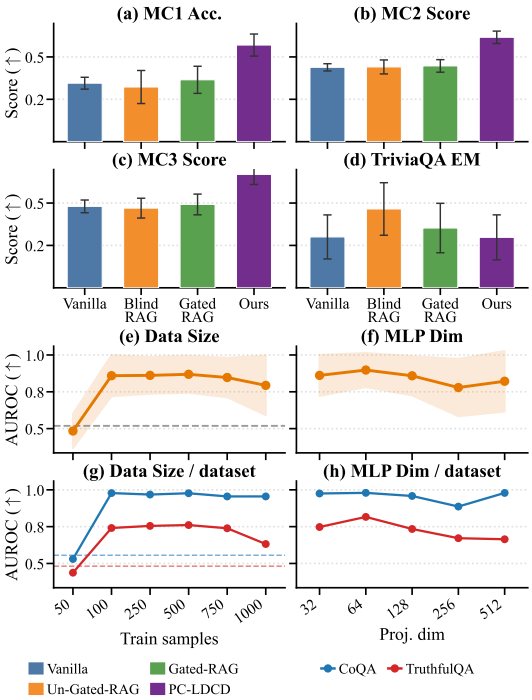

Figure 5: Results of the additional benchmark and ablations. **(a)**-**(d)** TruthfulQA MC1/MC2/MC3 and TriviaQA EM for vanilla, un-gated RAG, gated RAG, and **PC-LDCD** (bars indicate mean over Llama-3.2-1B and Mistral-7B; error bars indicate the corresponding std). **(e)**, **(g)**: PCNet detection AUROC on CoQA and TruthfulQA as a function of training-set size, while **(f)**, **(h)** as a function of projection dimension (top-right, bottom-right); dashed lines mark the Token NLL baseline. **(e)**, **(f)** represent the average across datasets and LLMs, with the shadow being the std, whilst **(g)** and **(h)** represent the avg. across LLMs.

5 Conclusions

We introduced **PCNet**, a tractable density estimator based on Probabilistic Circuits, and **PC-LDCD**, a density-gated contrastive decoding strategy that together resolve the Detection-Correction Asymmetry in LLM hallucination mitigation. By framing hallucinations as geometric anomalies on the factual manifold and computing exact NLL in a single forward pass, **PCNet** achieves near-perfect detection across CoQA, SQuAD v2.0, and TriviaQA (AUROC up to 0.99) without sampling, external verifiers, or weight modifications.

Crucially, using **PCNet** as a dynamic gate prevents indiscriminate corrections from corrupting factual generations, reducing the mean corruption rate to 53.7% and raising the preservation rate to 79.3%. **PC-LDCD** then routes interventions exclusively into the discrete token space, achieving the highest True+Info, MC2, and MC3 scores on TruthfulQA in three out of four models, while remaining competitive on standard QA benchmarks. Ablation studies confirm robustness to projection dimensionality and strong data efficiency, with near-optimal detection already at n=100 training samples. Qualitative correction examples are reported in Appendix E, while future directions and broader impacts are discussed respectively in Appendix F and Appendix G.

Limitations. **PCNet** requires a small calibration set of labeled factual and hallucinated hidden states, which may be costly in new domains. Moreover, our evaluation is limited to models up to 8B parameters, leaving open whether geometric separation holds in larger models. Finally, **PC-LDCD** gains are stronger on generative benchmarks than on exact-match tasks, suggesting it is better suited to open-ended generation than precise answers.

References

[1] Lei Huang, Weijiang Yu, Weitao Ma, Weihong Zhong, Zhangyin Feng, Haotian Wang, Qianglong Chen, Weihua Peng, Xiaocheng Feng, Bing Qin, et al. A survey on hallucination in large language models: Principles, taxonomy, challenges, and open questions. *ACM Transactions on Information Systems*, 43(2):1–55, 2025.

[2] Amos Azaria and Tom Mitchell. The Internal State of an LLM Knows When It's Lying. In Houda Bouamor, Juan Pino, and Kalika Bali, editors, *Findings of the Association for Computational Linguistics: EMNLP 2023*, pages 967–976, Singapore, December 2023. Association for Computational Linguistics. doi: 10.18653/v1/2023.findings-emnlp.68. URL https://aclanthology.org/2023.findings-emnlp.68/.

[3] Samuel Marks and Max Tegmark. The Geometry of Truth: Emergent Linear Structure in Large Language Model Representations of True/False Datasets, 2024. URL https://arxiv.org/abs/2310.06824.

[4] Kenneth Li, Oam Patel, Fernanda Viégas, Hanspeter Pfister, and Martin Wattenberg. Inference-Time Intervention: Eliciting Truthful Answers from a Language Model. *Advances in Neural Information Processing Systems*, 36:41451–41530, 2023. URL https://proceedings.neurips.cc/paper_files/paper/2023/hash/81b8390039b7302c909cb769f8b6cd93-Abstract-Conference.html.

[5] Kevin Meng, David Bau, Alex Andonian, and Yonatan Belinkov. Locating and Editing Factual Associations in GPT. *Advances in Neural Information Processing Systems*, 35: 17359–17372, 2022. URL https://proceedings.neurips.cc/paper_files/paper/2022/hash/6f1d43d5a82a37e89b0665b33bf3a182-Abstract-Conference.html.

[6] Shaolei Zhang, Tian Yu, and Yang Feng. TruthX: Alleviating Hallucinations by Editing Large Language Models in Truthful Space. In Lun-Wei Ku, Andre Martins, and Vivek Srikumar, editors, *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 8908–8949, Bangkok, Thailand, August 2024. Association for Computational Linguistics. doi: 10.18653/v1/2024.acl-long.483. URL https://aclanthology.org/2024.acl-long.483/.

[7] Hoifung Poon and Pedro Domingos. Sum-Product Networks: A New Deep Architecture. In *Proceedings of the 27th Conference on Uncertainty in Artificial Intelligence (UAI)*, pages 337–346, 2011.

[8] Pedro Zuidberg Dos Martires. Probabilistic Neural Circuits. *Proceedings of the AAAI Conference on Artificial Intelligence*, 38(15):17280–17289, 2024. ISSN 2374-3468. doi: 10.1609/aaai.v38i15.29675. URL https://ojs.aaai.org/index.php/AAAI/article/view/29675.

[9] Daniel Xie, Maxwell J. Jacobson, Adil Wazeer, Haiyan Wang, Xinghang Zhang, and Yexiang Xue. Reducing Hallucinations in LLM-based Scientific Literature Analysis Using Peer Context Outlier Detection, 2026. URL https://arxiv.org/abs/2604.01461.

[10] Albert Q. Jiang, Alexandre Sablayrolles, Arthur Mensch, Chris Bamford, Devendra Singh Chaplot, Diego de las Casas, Florian Bressand, Gianna Lengyel, Guillaume Lample, Lucile Saulnier, Lélio Renard Lavaud, Marie-Anne Lachaux, Pierre Stock, Teven Le Scao, Thibaut Lavril, Thomas Wang, Timothée Lacroix, and William El Sayed. Mistral 7B, 2023. URL https://arxiv.org/abs/2310.06825.

[11] Llama Team. The Llama 3 Herd of Models, 2024. URL https://arxiv.org/abs/2407.21783.

[12] Qwen3 Team. Qwen3 Technical Report, 2025. URL https://arxiv.org/abs/2505.09388.

[13] Siva Reddy, Danqi Chen, and Christopher D. Manning. CoQA: A Conversational Question Answering Challenge, 2019. URL https://arxiv.org/abs/1808.07042.

[14] Mandar Joshi, Eunsol Choi, Daniel Weld, and Luke Zettlemoyer. TriviaQA: A Large Scale Distantly Supervised Challenge Dataset for Reading Comprehension. In Regina Barzilay and Min-Yen Kan, editors, *Proceedings of the 55th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 1601–1611, Vancouver, Canada, July 2017. Association for Computational Linguistics. doi: 10.18653/v1/P17-1147. URL https://aclanthology.org/P17-1147/.

[15] Pranav Rajpurkar, Robin Jia, and Percy Liang. Know What You Don't Know: Unanswerable Questions for SQuAD, 2018. URL https://arxiv.org/abs/1806.03822.

[16] Stephanie Lin, Jacob Hilton, and Owain Evans. TruthfulQA: Measuring How Models Mimic Human Falsehoods, 2022. URL https://arxiv.org/abs/2109.07958.

[17] Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Küttler, Mike Lewis, Wen-tau Yih, Tim Rocktäschel, Sebastian Riedel, and Douwe Kiela. Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks. In H. Larochelle, M. Ranzato, R. Hadsell, M.F. Balcan, and H. Lin, editors, *Advances in Neural Information Processing Systems*, volume 33, pages 9459–9474. Curran Associates, Inc., 2020. URL https://proceedings.neurips.cc/paper_files/paper/2020/file/6b493230205f780e1bc26945df7481e5-Paper.pdf.

[18] Sebastian Farquhar, Jannik Kossen, Lorenz Kuhn, and Yarin Gal. Detecting hallucinations in large language models using semantic entropy. *Nature*, 630(8017):625–630, 2025. ISSN 1476-4687. doi: 10.1038/s41586-024-07421-0. URL https://www.nature.com/articles/s41586-024-07421-0.

[19] Jannik Kossen, Jiatong Han, Muhammed Razzak, Lisa Schut, Shreshth Malik, and Yarin Gal. Semantic entropy probes: Robust and cheap hallucination detection in llms. *arXiv preprint arXiv:2406.15927*, 2024.

[20] Xuefeng Du, Chaowei Xiao, and Yixuan Li. HaloScope: Harnessing Unlabeled LLM Generations for Hallucination Detection. *Advances in Neural Information Processing Systems*, 37:102948–102972, 2024. doi: 10.52202/079017-3270. URL https://proceedings.neurips.cc/paper_files/paper/2024/hash/ba92705991cfbbcedc26e27e833ebbae-Abstract-Conference.html.

[21] Robert Friel and Atindriyo Sanyal. Chainpoll: A high efficacy method for LLM hallucination detection, 2023. URL http://arxiv.org/abs/2310.18344.

[22] Sangwoo Heo, Sungwook Son, and Hyunwoo Park. HaluCheck: Explainable and verifiable automation for detecting hallucinations in LLM responses. *Expert Systems with Applications*, 272:126712, 2025. ISSN 0957-4174. doi: 10.1016/j.eswa.2025.126712. URL https://www.sciencedirect.com/science/article/pii/S0957417425003343.

[23] Weixuan Wang, Jingyuan Yang, and Wei Peng. Semantics-Adaptive Activation Intervention for LLMs via Dynamic Steering Vectors, 2025. URL https://arxiv.org/abs/2410.12299.

[24] Weixiang Zhao, Jiahe Guo, Yulin Hu, Yang Deng, An Zhang, Xingyu Sui, Xinyang Han, Yanyan Zhao, Bing Qin, Tat-Seng Chua, and Ting Liu. AdaSteer: Your Aligned LLM is Inherently an Adaptive Jailbreak Defender. In Christos Christodoulopoulos, Tanmoy Chakraborty, Carolyn Rose, and Violet Peng, editors, *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, pages 24559–24577, Suzhou, China, November 2025. Association for Computational Linguistics. ISBN 979-8-89176-332-6. doi: 10.18653/v1/2025.emnlp-main.1248. URL https://aclanthology.org/2025.emnlp-main.1248/.

[25] Kewei Liao, Tianbo Wang, Yuqing Ma, Zhange Zhang, Zhicheng Geng, Xiaowei Zhao, Jiakai Wang, and Xianglong Liu. Query-Routed Activation Editing with Truth-hierarchical Preference Optimization. *Proceedings of the AAAI Conference on Artificial Intelligence*, 40(38):31979–31987, 2026. ISSN 2374-3468. doi: 10.1609/aaai.v40i38.40468. URL https://ojs.aaai.org/index.php/AAAI/article/view/40468.

[26] Xiang Lisa Li, Ari Holtzman, Daniel Fried, Percy Liang, Jason Eisner, Tatsunori Hashimoto, Luke Zettlemoyer, and Mike Lewis. Contrastive Decoding: Open-ended Text Generation as Optimization. In Anna Rogers, Jordan Boyd-Graber, and Naoaki Okazaki, editors, *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 12286–12312, Toronto, Canada, July 2023. Association for Computational Linguistics. doi: 10.18653/v1/2023.acl-long.687. URL https://aclanthology.org/2023.acl-long.687/.

[27] Yung-Sung Chuang, Yujia Xie, Hongyin Luo, Yoon Kim, James Glass, and Pengcheng He. DoLa: Decoding by Contrasting Layers Improves Factuality in Large Language Models, 2024. URL https://arxiv.org/abs/2309.03883.

[28] Yue Zhang, Leyang Cui, Wei Bi, and Shuming Shi. Alleviating Hallucinations of Large Language Models through Induced Hallucinations. In Luis Chiruzzo, Alan Ritter, and Lu Wang, editors, *Findings of the Association for Computational Linguistics: NAACL 2025*, pages 8233–8247, Albuquerque, New Mexico, April 2025. Association for Computational Linguistics. ISBN 979-8-89176-195-7. doi: 10.18653/v1/2025.findings-naacl.459. URL https://aclanthology.org/2025.findings-naacl.459/.

[29] Adnan Darwiche. A Differential Approach to Inference in Bayesian Networks. *Journal of the ACM*, 50(3):280–305, 2003. doi: 10.1145/765568.765570.

[30] Wilson Hsu, Agastya Kalra, and Pascal Poupart. Online Structure Learning for Sum-Product Networks with Gaussian Leaves, 2017. URL http://arxiv.org/abs/1701.05265.

[31] Anji Liu, Stephan Mandt, and Guy Van den Broeck. Lossless Compression with Probabilistic Circuits, 2022. URL http://arxiv.org/abs/2111.11632.

[32] Weixin Chen, Simon Yu, Huajie Shao, Lui Sha, and Han Zhao. Neural Probabilistic Circuits: Enabling Compositional and Interpretable Predictions through Logical Reasoning, 2025. URL http://arxiv.org/abs/2501.07021.

[33] Jonathan Pan. Conversational Context Classification: A Representation Engineering Approach, 2026. URL https://arxiv.org/abs/2601.12286.

[34] Diederik P. Kingma and Jimmy Ba. Adam: A Method for Stochastic Optimization, 2017. URL https://arxiv.org/abs/1412.6980.

[35] Yejin Bang, Ziwei Ji, Alan Schelten, Anthony Hartshorn, Tara Fowler, Cheng Zhang, Nicola Cancedda, and Pascale Fung. HalluLens: LLM Hallucination Benchmark. *arXiv preprint arXiv:2504.17550*, 2025. URL https://arxiv.org/abs/2504.17550.

[36] YooJung Choi, Antonio Vergari, and Guy Van den Broeck. Probabilistic Circuits: A Unifying Framework for Tractable Probabilistic Models. Technical report, UCLA, 2020. URL http://starai.cs.ucla.edu/papers/ProbCirc20.pdf.

[37] Antonio Vergari, YooJung Choi, Anji Liu, Stefano Teso, and Guy Van den Broeck. A Compositional Atlas of Tractable Circuit Operations for Probabilistic Inference. In *Advances in Neural Information Processing Systems*, volume 34, 2021. URL https://proceedings.neurips.cc/paper_files/paper/2021/file/6e01383fd96a17ae51cc3e15447e7533-Paper.pdf.

[38] Tim Dettmers, Artidoro Pagnoni, Ari Holtzman, and Luke Zettlemoyer. QLoRA: Efficient Finetuning of Quantized LLMs. In A. Oh, T. Naumann, A. Globerson, K. Saenko, M. Hardt, and S. Levine, editors, *Advances in Neural Information Processing Systems*, volume 36, pages 10088–10115. Curran Associates, Inc., 2023. URL https://proceedings.neurips.cc/paper_files/paper/2023/file/1feb87871436031bdc0f2beaa62a049b-Paper-Conference.pdf.

[39] Stephen Robertson and Hugo Zaragoza. *The probabilistic relevance framework: BM25 and beyond*, volume 4. Now Publishers Inc, 2009.

A Theoretical foundations

We provide the theoretical grounding for **PCNET** and **PC-LDCD**. We refer the reader to [29] and [7] for foundational treatments of probabilistic circuits, including formal treatments of smoothness, decomposability, and the linear-time inference guarantees we leverage; Choi et al. [36] provides a modern unified framework, and Vergari et al. [37] systematizes tractable operations on circuits.

A.1 Consistency of the contrastive manifold estimator

Proposition 1 (Consistency) *Let $h \sim p_{factual}$ denote a factual hidden state. Assume there exists $\phi^\star$ such that the density of $f_{\phi^\star}(h)$ lies in the PC family, and that the family is identifiable. Then under standard regularity conditions, the minimizers $(\hat{\theta}_N, \hat{\phi}_N)$ of $\mathcal{L}(\theta, \phi)$ over N paired samples converge in probability to the population minimizer $(\theta^\star, \phi^\star)$ as $N \to \infty$ by standard M-estimator arguments.*

Remark 1 *The contrastive margin term acts as a bounded regularizer; under the stated assumptions, it does not alter the asymptotic consistency of the maximum-likelihood component, provided $\gamma > 0$ is held fixed.*

A.2 No-regret on confident states

Proposition 2 (No-regret on factual states) *Define the gating strength $\beta_t = \sigma(\mathcal{S}_{NLL}(z_t) - \tau)$. If $\mathcal{S}_{NLL}(z_t) \leq \tau - \log(1/\delta - 1)$, then $\beta_t \leq \delta$. In this regime, the score assigned to every candidate c_i in* (5) *satisfies*

$$\left| \mathrm{Score}_{LDCD}(c_i) - \log P_{LM}(c_i \mid x_{<t}) \right| \;\leq\; \delta \cdot \max_c |\mathcal{S}_{NLL}(f_\phi(h_{t+1}^{(c)}))| \tag{6}$$

so **PC-LDCD** *is δ-close to greedy decoding on states the* **PCNET** *deems factual. In particular, for $\delta \to 0$,* **PC-LDCD** *degenerates to the base LLM on high-confidence factual states, formally justifying the preservation rate guarantee.*

A.3 Per-token computational overhead

Proposition 3 (Per-token overhead) *At each generated token,* **PCNET** *incurs three additive costs: (i) a single MLP encoder pass of cost $O(D_{LLM} \cdot D_{PC})$; (ii) a single PC forward pass of cost $O(|\mathcal{N}|)$, where $|\mathcal{N}|$ is the number of nodes in the circuit (see Definition 1 and 29); (iii) for tokens flagged as anomalous only, k partial lookahead passes through the final transformer block to materialize $\{h_{t+1}^{(c_i)}\}_{i=1}^k$. The amortized per-token overhead is therefore*

$$O(D_{LLM} \cdot D_{PC} \;+\; |\mathcal{N}| \;+\; \mathrm{IGR} \cdot k \cdot d_{block}),$$

where d_{block} is the cost of one final-block evaluation. For our default configuration ($|\mathcal{N}| \approx 10^3$, $D_{LLM} = 4096$, $D_{PC} = 128$, $k = 8$, $\mathrm{IGR} \in [0.41, 0.78]$ across models), the PC scoring cost is below 2% of one full transformer-layer forward pass; the lookahead term dominates and scales as $\mathrm{IGR} \cdot k$ extra block evaluations per generated token.

The IGR values reported in our experiments (Table 3 and Table 5) thus reflect the actual lookahead frequency, providing an empirical proxy for the wall-clock overhead of **PC-LDCD**.

B Experimental setup

All experiments were conducted on a high-performance computing cluster using a single NVIDIA A100-SXM4 GPU with 40GB of VRAM, supported by 8 CPU cores and 35GB of system memory. Our implementation leverages `PyTorch` and `HuggingFace Transformers` within a `CUDA 12.5.0` environment. The target LLM is frozen and loaded in 4-bit NormalFloat (NF4) quantization [38] with FP16 compute dtype, ensuring memory efficiency without compromising the integrity of the latent representation manifolds extracted for **PCNET** density estimation.

The **PCNET** and its MLP bottleneck are trained jointly using the contrastive objective described in Section 3.2. Following [20], we reserve 25% of available QA pairs per dataset for testing. For

correction evaluation, performance is measured on a balanced subset of 300 samples (150 factual, 150 hallucinated), ensuring that corruption and preservation rates are computed on equal class support. All trials are run with three fixed RNG seeds (42, 43, 44) within a dedicated `Conda` environment. Full hyperparameters are reported in Table 4.

Table 4: Hyperparameters used for **PCNET** training across all experiments.

Hyperparameter	Value
Training samples	500 (250 factual / 250 hallucinated)
Epochs	50
Batch size	8
Learning rate	10^{-3}
Weight decay	10^{-5}
Loss weight α	0.8
Margin γ	5.0
Gradient clipping	$\|\nabla\|_2 \leq 1.0$
MLP projection dim d	128
PC depth	4
PC branching factor	3
LLM quantization	4-bit NF4 (FP16 compute)
RNG seeds	42, 43, 44

Computational budget The full experimental pipeline comprises 216 model-training experiments across five experiment groups: baselines (48), **PCNET** main runs (120), projection dimension ablations (16), RAG ablations (8), and data-size ablations (24). All jobs were allocated 48 GPU-hours, each on a single NVIDIA A100-SXM4 (40GB), yielding a theoretical upper bound of $216 \times 48 = 10368$ GPU-hours. Accounting for I/O overhead and average GPU utilization ($\sim$40%), the effective compute is estimated at approximately **4147** GPU-hours. The **PCNET** runs reported in the main text represent the largest share (5760 allocated GPU-hours).

B.1 Large Language Models, benchmark datasets, and evaluation metrics

Large Language Models To evaluate the robustness of our framework across varying architectures and parameter scales, we conduct experiments on four LLMs spanning 1B to 8B parameters: Llama-3.2-1B-Instruct [11], Qwen3-4B [12], Mistral-7B-v0.3 [10], and Llama-3.1-8B-Instruct [11]. This selection covers a broad range of model families, instruction-tuning regimes, and parameter scales, allowing us to assess whether **PCNET**'s geometric separation of factual and hallucinated hidden states transfers across architectures.

Benchmark Datasets We assess these models on four benchmarks targeting complementary factual challenges. **CoQA** [13] evaluates conversational reasoning over grounded passages, requiring the model to track dialogue context across turns. **TriviaQA** [14] targets knowledge-intensive open-domain QA, where correct answers depend on parametric world knowledge rather than in-context evidence. **SQuAD v2.0** [15] tests reading comprehension with unanswerable questions, probing the model's ability to abstain rather than hallucinate an answer. Finally, **TruthfulQA** [16] specifically targets questions designed to elicit misconceptions encoded in pretraining weights, making it the most adversarial benchmark in our suite.

Evaluation Metrics We evaluate across three axes. For **detection**, we report AUROC and F1, measuring the ability of **PCNET** to separate hallucinated from factual hidden states at the instance level. For **generation quality**, we use Exact Match (EM) on TriviaQA and SQuAD v2.0, and token-level F1 on CoQA. On TruthfulQA, we report True+Info (T+I), MC1, MC2, and MC3: T+I jointly measures factual grounding and informativeness; MC1 measures single-answer accuracy against the most likely misconception; MC2 reports calibrated probability mass over all correct completions; and MC3 measures likelihood ranking against distractors. MC2 and MC3 are particularly diagnostic as they capture distributional truthfulness rather than single-answer confidence. Finally, we report the **Instance Gating Rate** (IGR), i.e., the fraction of decoding steps where **PCNET** triggers an intervention, alongside the **Corruption** rate (fraction of originally correct generations degraded by intervention) and the **Preservation** rate (fraction of correct generations successfully protected by gating), to jointly assess intervention selectivity and the safety of the gating mechanism. In Table 3 and Table 5, the IGR is identical across all correction methods within the same model; as all

approaches share the same **PCNet** gating signal, intervention decisions are made solely by **PCNet**, independently of the downstream correction strategy applied.

C Detailed results

Table 5 reports the full correction performance on the standard QA benchmarks (CoQA, SQuAD v2.0, TriviaQA) in terms of token-level F1 and Exact Match (EM). Results are formatted as *post-intervention score* (Δ), where Δ denotes the change relative to the vanilla baseline, and are averaged across three RNG seeds. The IGR reflects the fraction of prompts where **PCNet** actively triggered an intervention; a lower IGR indicates more selective gating, while a higher IGR reflects more pervasive anomaly detection on a given model–benchmark combination.

Table 5: Performance on standard QA Benchmarks (CoQA, SQuAD v2.0, TriviaQA). Results are formatted as $score_{\pm \text{std}}$ (Δ), where Δ denotes the change relative to Vanilla. IGR (%) reflects the average Instance Gating Rate. Best post-intervention scores per model are highlighted in **bold**. Averaged across three seeds.

Model	Method	IGR (%) ↓	CoQA F1 ↑	SQuAD v2.0 EM ↑	TriviaQA EM ↑
Llama-3.2-1B	Vanilla	0.0	0.50 (+0.00)	0.50 (+0.00)	0.50 (+0.00)
	DoLa	$77.9_{\pm 15.4}$	$0.68_{\pm .01}$ (+0.18)	$0.83_{\pm .00}$ (+0.33)	$0.58_{\pm .02}$ (+0.08)
	ITI	$77.9_{\pm 15.4}$	$0.48_{\pm .00}$ (-0.02)	$0.49_{\pm .01}$ (-0.01)	$0.47_{\pm .01}$ (-0.03)
	AdaSteer	$77.9_{\pm 15.4}$	$0.70_{\pm .02}$ (+0.20)	$\mathbf{0.86_{\pm .01}}$ **(+0.36)**	$\mathbf{0.63_{\pm .00}}$ **(+0.13)**
	SADI	$77.9_{\pm 15.4}$	$0.70_{\pm .02}$ (+0.20)	$\mathbf{0.86_{\pm .02}}$ **(+0.36)**	$\mathbf{0.63_{\pm .00}}$ **(+0.13)**
	ICD	$77.9_{\pm 15.4}$	$\mathbf{0.71_{\pm .01}}$ **(+0.21)**	$0.83_{\pm .01}$ (+0.33)	$0.59_{\pm .01}$ (+0.09)
	PC-LDCD (Ours)	$77.9_{\pm 15.4}$	$0.69_{\pm .02}$ (+0.19)	$0.84_{\pm .01}$ (+0.34)	$0.60_{\pm .01}$ (+0.10)
Qwen3-4B	Vanilla	0.0	0.50 (+0.00)	0.50 (+0.00)	0.50 (+0.00)
	DoLa	$45.4_{\pm 3.9}$	$0.56_{\pm .02}$ (+0.06)	$\mathbf{0.90_{\pm .03}}$ **(+0.41)**	$\mathbf{0.70_{\pm .02}}$ **(+0.20)**
	ITI	$45.4_{\pm 3.9}$	$0.53_{\pm .04}$ (+0.03)	$0.73_{\pm .12}$ (+0.23)	$0.51_{\pm .01}$ (+0.01)
	AdaSteer	$45.4_{\pm 3.9}$	$\mathbf{0.58_{\pm .02}}$ **(+0.08)**	$\mathbf{0.90_{\pm .02}}$ **(+0.41)**	$0.69_{\pm .02}$ (+0.19)
	SADI	$45.4_{\pm 3.9}$	$\mathbf{0.58_{\pm .02}}$ **(+0.08)**	$\mathbf{0.90_{\pm .03}}$ **(+0.41)**	$\mathbf{0.70_{\pm .02}}$ **(+0.20)**
	ICD	$45.4_{\pm 3.9}$	$\mathbf{0.58_{\pm .02}}$ **(+0.08)**	$\mathbf{0.90_{\pm .02}}$ **(+0.40)**	$0.67_{\pm .02}$ (+0.17)
	PC-LDCD (Ours)	$45.4_{\pm 3.9}$	$\mathbf{0.58_{\pm .02}}$ **(+0.08)**	$\mathbf{0.90_{\pm .03}}$ **(+0.40)**	$0.67_{\pm .01}$ (+0.17)
Mistral-7B	Vanilla	0.0	0.50 (+0.00)	0.50 (+0.00)	0.50 (+0.00)
	DoLa	$58.3_{\pm 6.0}$	$0.71_{\pm .01}$ (+0.21)	$0.84_{\pm .03}$ (+0.34)	$0.82_{\pm .01}$ (+0.32)
	ITI	$58.3_{\pm 6.0}$	$0.53_{\pm .02}$ (+0.03)	$0.50_{\pm .01}$ (+0.00)	$0.50_{\pm .00}$ (+0.00)
	AdaSteer	$58.3_{\pm 6.0}$	$0.70_{\pm .01}$ (+0.20)	$\mathbf{0.86_{\pm .02}}$ **(+0.36)**	$\mathbf{0.84_{\pm .01}}$ **(+0.34)**
	SADI	$58.3_{\pm 6.0}$	$0.71_{\pm .01}$ (+0.21)	$\mathbf{0.86_{\pm .02}}$ **(+0.36)**	$\mathbf{0.84_{\pm .01}}$ **(+0.34)**
	ICD	$58.3_{\pm 6.0}$	$0.68_{\pm .01}$ (+0.18)	$0.67_{\pm .02}$ (+0.17)	$0.78_{\pm .02}$ (+0.28)
	PC-LDCD (Ours)	$58.3_{\pm 6.0}$	$\mathbf{0.73_{\pm .01}}$ **(+0.23)**	$0.85_{\pm .02}$ (+0.35)	$0.81_{\pm .01}$ (+0.31)
Llama-3.1-8B	Vanilla	0.0	0.50 (+0.00)	0.50 (+0.00)	0.50 (+0.00)
	DoLa	$41.4_{\pm 5.6}$	$0.72_{\pm .00}$ (+0.22)	$0.87_{\pm .02}$ (+0.37)	$0.79_{\pm .00}$ (+0.29)
	ITI	$41.4_{\pm 5.6}$	$0.50_{\pm .01}$ (+0.00)	$0.50_{\pm .00}$ (+0.00)	$0.50_{\pm .00}$ (+0.00)
	AdaSteer	$41.4_{\pm 5.6}$	$0.70_{\pm .00}$ (+0.20)	$0.91_{\pm .02}$ (+0.41)	$\mathbf{0.82_{\pm .00}}$ **(+0.32)**
	SADI	$41.4_{\pm 5.6}$	$0.71_{\pm .01}$ (+0.21)	$\mathbf{0.92_{\pm .02}}$ **(+0.42)**	$\mathbf{0.82_{\pm .00}}$ **(+0.32)**
	ICD	$41.4_{\pm 5.6}$	$\mathbf{0.76_{\pm .01}}$ **(+0.26)**	$0.83_{\pm .02}$ (+0.33)	$0.76_{\pm .01}$ (+0.26)
	PC-LDCD (Ours)	$41.4_{\pm 5.6}$	$0.69_{\pm .01}$ (+0.19)	$0.90_{\pm .03}$ (+0.40)	$0.79_{\pm .01}$ (+0.29)

D Additional benchmark and ablations

D.1 Benchmark against RAG-augmented generation

The RAG ablation evaluates four approaches: *Vanilla* (base LLM, no intervention), **PC-LDCD** (PCNet-gated correction), *Un-Gated RAG* (retrieval always applied), and *Gated RAG* (retrieval triggered only when **PCNet** flags an anomaly). Retrieval is implemented via BM25 (Okapi, k_1=1.5, b=0.75) [39] over a corpus of up to 10000 TriviaQA training passages, with top-k=3 passages pre-appended to the prompt as in-context evidence. For TruthfulQA-MC, all MC1 and MC2 choices are scored via teacher-forced mean log-probability; under Gated RAG, the retrieval-augmented prompt replaces the vanilla prompt only for flagged samples. The anomaly threshold τ is calibrated per model on the NLL distribution of the dataset's stored answers, selecting the value that maximizes the F1 score on a held-out validation split. For MC scoring, where ground-truth labels are unavailable at inference time, the median NLL across all choices is used as the threshold. All arms share the same

frozen LLM and PCNET checkpoint; only the prompt construction and decoding strategies differ across conditions.

D.2 Ablation on training data size

The data-size ablation trains PCNET with $n \in \{50, 100, 250, 500, 750, 1000\}$ balanced samples (50% factual / 50% hallucinated). For each size condition, the MLP projector and PCNET are reinitialized from scratch while the base LLM remains frozen and is loaded only once per model–dataset pair. All other hyperparameters (depth= 4, d=128, 50 epochs, lr= 10^{-3}, margin= 5.0, α=0.8) are held fixed. Detection is evaluated on a held-out balanced test set of 200 samples using AUROC and F1. The results are reported in Figure 6.

D.3 Ablation on MLP projection dimension

The projection-dimension ablation sweeps $d \in \{32, 64, 128, 256, 512\}$ while keeping all other hyperparameters fixed as in Appendix B. For each value of d, the MLP projector and PCNET are reinitialized independently. Detection performance is measured via AUROC; latent alignment quality is assessed via mean cosine similarity and the fraction of positively-aligned dimensions between the correction direction and the factual manifold, computed over held-out hidden states. The results are reported in Figure 7.

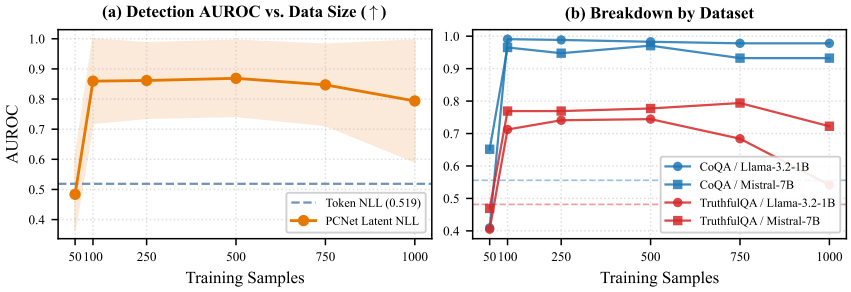

Figure 6: AUROC detection achieved by PCNET on Llama 3.2-1B and Mistral-7B LLMs and on CoQa and TruthfulQA benchmark settings across different training dataset sizes. **(a)** The line represents the average across LLMs and datasets, and the shadow represents the standard deviation. **(b)** Each line refers to a single execution.

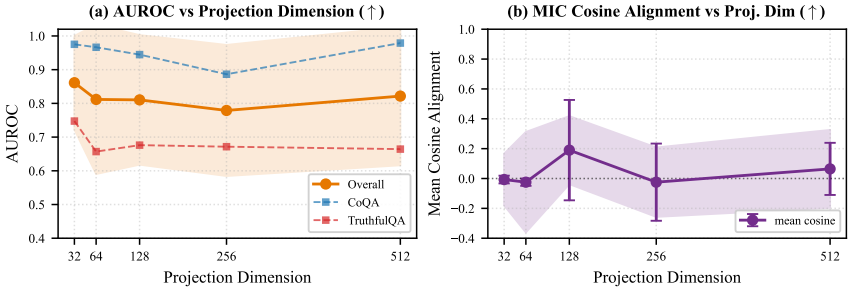

Figure 7: MLP projection dimensionality ablation across $d \in \{32, 64, 128, 256, 512\}$ on Llama-3.2-1B and Mistral-7B over CoQA and TruthfulQA. **(a)** Mean AUROC across all settings. **(b)** Latent alignment metrics (mean cosine similarity and fraction of positively-aligned dimensions), where d=128 achieves the highest alignment. Lines represent the mean across all settings, with shaded areas and error bars denoting the standard deviation.

E Qualitative examples of PC-LDCD gated correction

We present comprehensive qualitative examples of our gated correction mechanism in action. The examples are categorized by whether the model correctly detected and fixed a hallucination (Successfully Corrected), failed to fix it (Correction Failed), successfully preserved a correct answer (Not Corrected - Preserved), or incorrectly intervened (Corrupted).

Example 1 — Mistral-7B on *TriviaQA*

- **Context**: Who was Henry VIII's second wife?
- **Answer**: PARIS
- **Correction**: Anne Boleyn. *(Successfully Corrected)*

Example 2 — Llama-3.1-8B on *TriviaQA*

- **Context**: Which other car company owns Seat?
- **Answer**: Borgia
- **Correction**: Volkswagen Group owns Seat. *(Successfully Corrected)*

Example 4 — Qwen3-4B on *TriviaQA*

- **Context**: What is thalassophobia a fear of?
- **Answer**: Jonathan Swift
- **Correction**: Thalassophobia is a fear of the sea. It is a specific phobia characterized by an intense and irrational fear of oceans, water, or the sea. People with thalassophobia may experience anxiety, panic attacks, or avoidance behaviors when near the ocean or water. This fear can stem from various psychological, cultural, or personal experiences. *(Successfully Corrected)*

Example 5 — Mistral-7B on *TriviaQA*

- **Context**: What is the nickname of Sheffield United
- **Answer**: Hugh Quarshie
- **Correction**: United *(Correction Failed)*

Example 6 — Qwen3-4B on *TriviaQA*

- **Context**: Brown Willy is the highest point in which county?
- **Answer**: ROBERT SCHUMANN
- **Correction**: Brown Willy is the highest point in the county of Cornwall, England. It is a hill located in the south-western part of England, specifically in the area known as the Dartmoor region. The hill stands at an elevation of approximately 290 meters (951 feet) above sea level. It is part of the Dartmoor National Park, which is known for its *(Correction Failed)*

Example 7 — Llama-3.2-1B on *TriviaQA*

- **Context**: Who led the quest for the golden fleece
- **Answer**: Rome
- **Correction**: None of them. The quest for the golden fleece was led by Philip II of Macedon. He was a king of Macedon in Greece. He led a series of expeditions to search for it. The story of the quest for the golden fleece is from the Iliad by Homer. It is a part of the epic poem that tells the story of the Greeks and their battles against the Tro *(Correction Failed)*

Example 9 — Llama-3.2-1B on *TriviaQA*

- **Context**: What is thalassophobia a fear of?
- **Answer**: The Sea
- **Correction**: The Sea *(Not Corrected - Preserved)*

Example 10 — Mistral-7B on *TriviaQA*

- **Context**: What is thalassophobia a fear of?
- **Answer**: The Sea
- **Correction**: The Sea *(Not Corrected - Preserved)*

Example 11 — Llama-3.2-1B on *TriviaQA*

- **Context**: Who was Henry VIII's second wife?
- **Answer**: Anne Boleyn
- **Correction**: Anne Boleyn *(Not Corrected - Preserved)*

Example 12 — Llama-3.1-8B on *TriviaQA*

- **Context**: What is the county town of Mayo?
- **Answer**: CASTLE BAR
- **Correction**: CASTLE BAR *(Not Corrected - Preserved)*

Example 13 — Qwen3-4B on *TriviaQA*

- **Context**: Botswana was known by what name until 1966?
- **Answer**: Bechuanaland
- **Correction**: Before 1966, Botswana was known as Bechuanaland. The region was under British administration, and it was referred to as Bechuanaland Protectorate. In 1966, following a referendum and independence movement led by the Batswana people, the territory officially became the Republic of Botswana. The name change from Bechuanaland to Botswana *(Corrupted)*

Example 14 — Mistral-7B on *TriviaQA*

- **Context**: What is the county town of Mayo?
- **Answer**: CASTLE BAR
- **Correction**: Castlebar is the county town of County Mayo in Ireland. It is located in the west of Ireland and is the largest town in County Mayo. Castlebar is known for its vibrant culture, historical sites, and natural beauty. It is also a hub for commerce, education, and entertainment in the region. The town was founded in the late 18th century and has a rich *(Corrupted)*

Example 15 — Mistral-7B on *TriviaQA*

- **Context**: Sometimes called the 'Invicta flag', which English county's flag is a white horse on a red background?
- **Answer**: KENT
- **Correction**: Hampshire's flag is known as the 'Invicta flag' and features a horse on red background. The flag is based on the arms granted to Hampshire in 1660, which depict a horse rampant. The name 'Invicta' comes from the Latin word meaning 'unconquered', and refers to Hampshire's status as one of only two counties *(Corrupted)*

F Future works

Several promising directions emerge from this work. First, our framework currently extracts the hidden state of the *last* token as the density input; however, recent evidence suggests that early token representations may encode factual commitments before the generation unfolds [3]. A rigorous investigation of whether the *first* latent token already carries sufficient factual signal—and whether this enables earlier, cheaper anomaly detection—could substantially reduce inference overhead while improving robustness.

Second, and most consequentially, the advent of long chain-of-thought reasoning models (e.g., DeepSeek-R1, o1-style models) introduces a structured intermediate space that our framework has not yet exploited. We conjecture that individual *reasoning steps* in the chain-of-thought trace can be independently scored by **PCNet**, enabling the identification and targeted correction of the precise step where a hallucinated factual commitment first emerges—rather than correcting the final answer after the error has propagated. This would transform **PC-LDCD** from a token-level intervention into a *step-level reasoning corrector*, with the potential to dramatically improve factual consistency in mathematical, scientific, and multi-hop reasoning tasks where error accumulation across steps is the primary failure mode.

Third, our ablation against RAG reveals a complementary performance profile: **PC-LDCD** achieves superior distributional truthfulness while RAG excels at knowledge-lookup tasks with explicit retrieved evidence. A natural extension is therefore a hybrid architecture of the two approaches.

Finally, scaling our evaluation to models beyond 8B parameters and extending **PCNet** to multilingual and multimodal residual streams remain important open directions for establishing the broader applicability of latent density estimation as a foundation for trustworthy LLM inference.

G Broader impact

By improving the factual reliability of LLMs, our framework contributes positively to the deployment of trustworthy AI systems in high-stakes domains, potentially reducing the spread of generated misinformation. However, as with any representation engineering technique, there is a risk of dual-use: the gating and manifold-learning mechanisms could theoretically be manipulated to enforce a biased or ideologically skewed "factual" manifold. Future work must ensure that the calibration datasets used for density estimation are diverse, transparent, and rigorously evaluated for representation biases.